POWER WINDOW CONTROL MODULE/POWER OUTER MIRROR CONTROL MODULE

id091200014900

Purpose

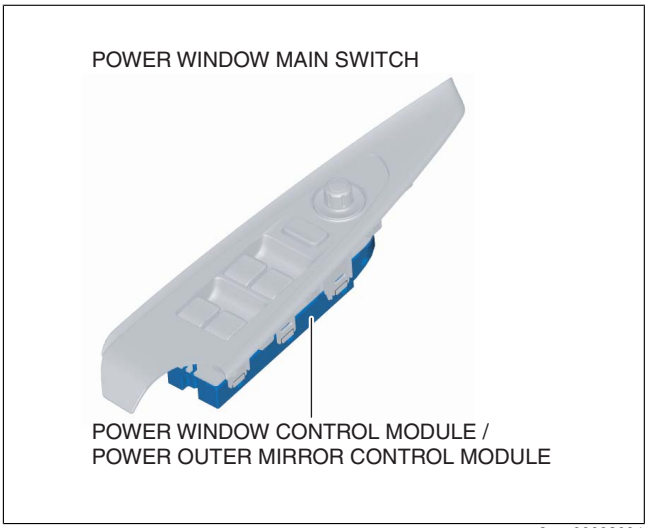
- Controls the driver's side power window system based on the power window main switch operation signal and power window motor (driver's side) pulse.
- Controls the power outer mirror system based on the power window man switch operation signal.

Function

- When the power window control module receives a power window main switch operation signal, it turns on/off the open/close relay.
- The power window control module detects the power window motor operation status based on the pulse from the power window motor (driver's side).

Construction

- The power window control module and the power outer mirror control module are built into the power window main switch.

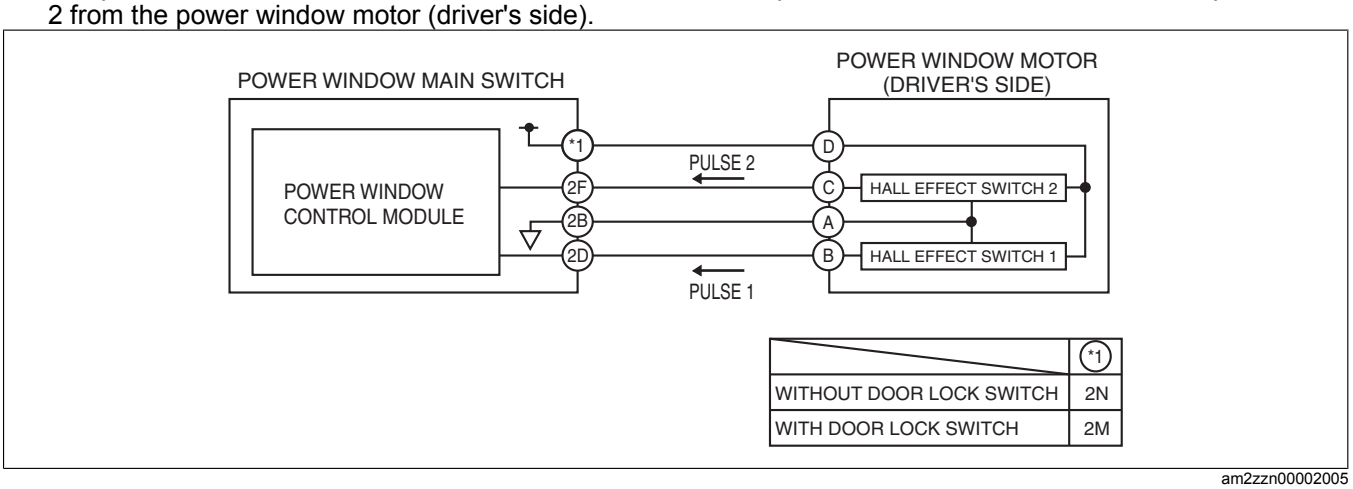


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Operation

Power window control module

- The power window control module detects the motor rotation speed and rotation direction based on pulse 1 and 2 from the power window motor (driver's side).



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WITHOUT DOOR LOCK SWITCH	2N
WITH DOOR LOCK SWITCH	2M

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Power outer mirror control module

- For the power outer mirror control module operation, refer to [POWER OUTER MIRROR SYSTEM]. (See POWER OUTER MIRROR SYSTEM.)

Fail-safe

- When power window control module detects a pulse (Hall effect switch) error, it switches the power window system to fail-safe mode to prevent incorrect operation.

Malfunction condition	Ignition is switched ON (engine off or on)	Ignition is switched off	Recovery condition
Power window control module detected pulse 1 error (for pinching and door glass position detection) stop malfunction • During close or open operation, pulse 1 cannot be detected while pulse 2 is detected	Auto operation is prohibited (Manual operation is enabled) Operation of power window main switch except for the driver's seat is prohibited	Auto and manual operations are prohibited	From a close operation, the power window system returns to normal mode when pulse 1 and 2 signals are received normally, and when the window fully-closed position and the window glass lowering amount during auto reverse are detected.
Power window control module detected pulse 2 error (for door glass direction detection) stop malfunction • During close or open operation, pulse 2 cannot be detected while pulse 1 is detected			
Power window control module detected pulse signal error (inversion of pulse 1 and pulse 2 input signals or large phase deviation between pulses 1 and 2) • Difference detected between the direction detected by the signals of pulses 1 and 2, and the actual direction during close or open operation			
Power window control module detected pulse 1 error (Non-return range lowering) • During close operation, pulse 1 signal from the power window motor detected more than set times even after the window reached the fully-closed position recorded by the power window control module.			
Power window control module detected pulse 1 and 2 stop malfunction • Pulse signals 1 and 2 are not detected for 1 s after the open operation is started from the fully closed position			